INTERPRETIVE SUMMARIES, JULY 2023

Invited review: **From heat stress to disease—Immune response and candidate genes involved in cattle thermotolerance.** *By Lemal et al., page 4471.* Heat stress implies effects on immune system mechanisms in cattle. Thus, we explored the molecular mechanisms to better understand how heat stress could lead to a greater occurrence of immune-related diseases. We also associated these mechanisms with candidate genes involved in thermotolerance, and accordingly, we inferred potential molecular biomarkers for heat stress. The review of these findings contributes to enhanced phenotyping approaches, especially the use of immune markers, and ultimately, improved detection of heat stress in cattle.
https://doi.org/10.3168/jds.2022-22727.

Differential behavior of *Lactobacillus helveticus* B1929 and ATCC 15009 on the hydrolysis and angiotensin-I-converting enzyme inhibition activity of fermented ultra-high-temperature milk and nonfat dried milk powder. *By Guron et al., page 4502.* Fermented foods are receiving growing interest due to their reported health benefits. Research has shown that some milk proteins may alleviate specific conditions, such as hypertension. These properties are often encrypted in the native proteins and require liberation in vivo by gastrointestinal digestion or fermentation with proteolytic starter cultures (lactic acid bacteria). We fermented ultra-high-temperature milk and reconstituted nonfat dried milk, 2 commercially sourced products, with 2 *Lactobacillus helveticus* strains, B1929 and ATCC 15009, and examined their ability to develop antihypertensive activities of the fermentate. We showed that the B1929 performed better than ATCC 15009 to hydrolyze proteins and develop potentially advantageous properties.
https://doi.org/10.3168/jds.2022-22842.

In vitro gastric digestion of an experimental infant formula containing both intact and hydrolyzed milk proteins. *By Lambers et al., page 4524.* This study describes the digestion of an experimental infant formula containing both hydrolyzed and intact milk proteins. Observed in vitro digestive benefits compared with an intact milk protein formula suggest that replacing a proportion of the protein source of infant formula with a hydrolysate could result in a digestive benefit comparable to that of full hydrolysate formula, potentially benefiting from the bioactivity of intact milk proteins and the digestive benefit of hydrolysates. Overall, this may highlight a novel, innovative solution for infant formula to better align with the composition of human milk, which naturally contains hydrolyzed human milk proteins.
https://doi.org/10.3168/jds.2022-22667.

Growth performance, blood metabolites, ruminal fermentation, and bacterial community in preweaning dairy calves fed corn silage–included starter and total mixed ration. *By Zhang et al., page 4545.* Calf starter including whole-plant corn silage and whole-plant corn silage–based total mixed ration are both thought to improve rumen environment and calf performance. Despite the different characteristics of these diets, our results suggest that total mixed ration feeding has benefits in fiber intake, rumen fermentation, and relative abundance of ruminal cellulolytic bacteria without adverse effects on dry matter intake, blood metabolites, and growth performance. However, the starter including whole-plant corn silage had adverse effects on rumen fermentation parameters and the cellulolytic bacteria community. In conclusion, total mixed ration, but not the starter including whole-plant corn silage, has the potential to be an alternative starter in preweaning calves.
https://doi.org/10.3168/jds.2022-22476.

Rumen-protected choline and methionine during the periparturient period affect choline metabolites, amino acids, and hepatic expression of genes associated with one-carbon and lipid metabolism. *By Potts et al., page 4559.* Both choline and methionine have been shown to elicit changes in production when supplemented between late pregnancy and early lactation. In this study, supplemental choline consistently increased secretion of total lysophosphatidylcholine in milk for primi- and multiparous cows, although this effect was dependent on methionine supplementation in primiparous cows. Choline and methionine both altered secretion of several individual species of phosphatidylcholine in milk for multiparous cows and their concentrations in plasma for primiparous cows. Results suggest that both choline and methionine affect choline metabolism, but in different ways.
https://doi.org/10.3168/jds.2022-22334.

Effects on rumen pH and feed intake of a dietary concentrate challenge in cows fed rations containing pH modulators with different neutralizing capacity. *By Bach et al., page 4580.* In an attempt to meet the energy demands of high milk yields, the amount of nonfiber carbohydrates in the diet is increased and the supply of forage is reduced, which may lead to reductions in rumen pH and alter

the rumen microbiota and overall animal performance. Supplementing rations with neutralizing agents is a common strategy to minimize ruminal disruptions. Herein, the effects of diets supplemented with sodium bicarbonate or a magnesium-based product (pHix-Up, Timab Magnesium) on rumen pH, rumen microbial population, and milk performance of dairy cattle exposed to an induced decrease in rumen pH through a dietary challenge were evaluated. Cows supplemented with ~62 grams per day of magnesium oxide maintained a greater rumen pH and consumed more dry matter than cows supplemented with ~200 grams per day of sodium bicarbonate when fed a diet with a low forage-to-concentrate ratio.
https://doi.org/10.3168/jds.2022-22734.

Effects of medium-chain fatty acids and tributyrin supplementation in milk replacers on growth performance, blood metabolites, and hormone concentrations in Holstein dairy calves. *By Murayama et al., page 4599.* Specific fatty acids and their triglycerides affect growth, internal secretions, and gastrointestinal tract development. The effects of triglycerides containing medium-chain fatty acids and tributyrin supplementation in a milk replacer on growth performance, plasma metabolites, and hormone concentrations in calves were evaluated. This study showed that medium-chain fatty acids and tributyrin supplementation in a milk replacer might improve dairy calves' growth performance and gut health.
https://doi.org/10.3168/jds.2022-22957.

Effect of a garlic and citrus extract supplement on performance, rumen fermentation, methane production, and rumen microbiome of dairy cows. *By Khurana et al., page 4608.* Methane from enteric fermentation in ruminants is a main contributor to greenhouse gas emissions; feed supplements have the potential to reduce enteric methane production. Thus, the potential effects of a garlic and citrus extract–based natural feed supplement on methane emissions, rumen microbiome, and performance of dairy cows were evaluated. Enteric methane emissions from garlic and citrus extract–supplemented cows were lower, with no effect on milk performance. Microbiome richness and diversity did not differ between the 2 treatments. The higher relative abundance of *Succinivibrionaceae* family and a lower abundance of *Methanobrevibacter* in the supplemented group could be associated with reduced methane production.
https://doi.org/10.3168/jds.2022-22838.

Evaluation of 3 northwest European seaweed species on enteric methane production and lactational performance of Holstein-Friesian dairy cows. *By Muizelaar et al., page 4622.* There is considerable interest in seaweeds for their potential to reduce enteric methane emissions in cattle. In the present study, the red seaweed *Chondrus crispus*, the brown seaweed *Saccharina latissima*, and a mixture of the brown seaweeds *Fucus serratus* and *S. latissima* fed at 150 grams per day did not decrease methane emissions of dairy cattle and did not affect carbon dioxide or hydrogen emissions. Feed intake was not affected by these seaweeds, whereas *S. latissima* increased milk yield, fat- and protein-corrected milk yield, milk lactose content, and lactose yield of lactating dairy cattle.
https://doi.org/10.3168/jds.2022-22749.

Effect of feeding a *Saccharomyces cerevisiae* fermentation product to Holstein cows exposed to high temperature and humidity conditions on milk production performance and efficiency—A pen-level trial. *By Thomas et al., page 4650.* Heat stress and its detrimental effects on milk production efficiency are major concerns of the dairy industry. The objective of this study was to evaluate the effect of feeding a postbiotic product from yeast fermentation on milk production efficiency of Holstein cows exposed to high temperature and humidity conditions. The study was conducted in 2 commercial dairy farms in Mexico during a period of extreme environmental temperatures. Multiparous cows fed the *Saccharomyces cerevisiae* fermentation product (SCFP) produced approximately 1 kg more milk than control cows. Cows supplemented with the SCFP had greater feed efficiency and improved body condition in both primiparous and multiparous cows. Under these trial conditions, supplementation with the SCFP demonstrated advantages in the efficiency of milk production.
https://doi.org/10.3168/jds.2022-22516.

The effects of feeding α-amylase-enhanced corn silage with different dietary starch concentrations to lactating dairy cows on milk production, nutrient digestibility, and blood metabolites. *By Krogstad and Bradford, page 4666.* Corn silage is an important feed for dairy cattle, and improving its digestibility through crop selection may improve dairy production efficiency. Nutrient digestibility of α-amylase-enhanced corn silage was greater than the control early after harvest, but the differences waned with time. Across 2 different concentrations of starch, α-amylase-enhanced corn silage did not affect milk production and blood metabolites. Increasing dietary starch content increased energy-corrected milk and milk protein yield without affecting markers of systemic inflammation or metabolism.
https://doi.org/10.3168/jds.2022-23030.

Body and milk production traits as indicators of energy requirements and efficiency of purebred Holstein and 3-breed rotational crossbred cows from Viking Red, Montbéliarde, and Holstein sires. *By Piazza et al., page 4966.* Crossbreeding has been proposed to counterbalance the decline of longevity of purebred Holstein cows, but the lower milk yield of crossbred cows compared with Holsteins discourages its application in many farms. In this study, we compared crossbred cows obtained from a rotational scheme involving 3 breeds with purebred Holsteins for several body measurements and milk production metrics. Results confirmed that crossbreds yielded less milk than Holsteins. However, differences mostly disappeared when production traits were scaled on traits related to the energy costs of production, suggesting that efficiency measures are more effective in comparing crossbreds and purebreds than mere milk yields.
https://doi.org/10.3168/jds.2022-22830.

Modeling the effects of heat stress in animal performance and enteric methane emissions in lactating dairy cows. *By Souza et al., page 4725.* This article focuses on developing models that predict the changes in responses that are of economic and environmental interest with a cyclical heat stress period in lactating dairy cows. The predictions from the models generated in this study may help dairy nutritionists decide when and how to apply available strategies to attenuate the side effects of heat stress exposure on intake and milk yield. They can also help increase understanding of the effects of heat stress on enteric methane emissions and free water intake in lactating dairy cows.
https://doi.org/10.3168/jds.2022-22658.

Comparing thermal conditions inside and outside lactating dairy cattle barns in Canada. *By VanderZaag et al., page 4738.* This study measured confined dairy herds' exposure to heat stress conditions in a wide range of Canadian ecoregions. Equations were developed to predict in-barn conditions based on outdoor conditions by barn ventilation type. We gained a better understanding of patterns of in-barn temperature-humidity index exceedance compared with that of the outdoors and when increased ventilation-driven cooling might be beneficial. The suitability of climate data sources for heat stress studies was assessed, and the importance of identifying appropriate temperature-humidity index thresholds was highlighted. Evidence-based thresholds can directly affect the profitability of milk production by optimizing barn design and operating costs while minimizing heat stress–related economic losses.
https://doi.org/10.3168/jds.2022-22870.

Assessment of the current performance of grazing infrastructure across Irish dairy farms. *By Maher et al., page 4759.* In rotational grazing systems, pasture allocation directly affects grass utilization and herd productivity. Animal movement on farm roadway networks between the milking parlor and grazing paddocks imposes high labor costs and is a potential stressor for the herd. The infrastructure to support efficient grazing systems has developed through a legacy of historical management practices and is constrained by farm layout. Limited research exists regarding the current performance of grazing infrastructure and the identification of factors that affect grazing infrastructure performance. This paper aims to quantify the current performance of grazing infrastructure on dairy farms and roadway network efficiency relative to herd size.
https://doi.org/10.3168/jds.2022-22799.

Economic and environmental analysis of processing plant interventions to reduce fluid milk waste. *By Lau et al., page 4773.* Dairy product food loss and waste have a substantial economic and environmental impact. An analysis of 5 different processing plant interventions was conducted to determine the private and social gains of extending fluid milk shelf life. Increasing periodic equipment cleaning was shown to be the most cost-effective strategy to improve fluid milk shelf life. This analysis represents a method that processors can use to evaluate the value of an additional day of shelf life and different interventions in their processing plant.
https://doi.org/10.3168/jds.2022-23019.

Dung treated by high-temperature composting is an optimal bedding material for suckling calves according to analyses of microbial composition, growth performance, health status, and behavior. *By Zhou et al., page 4785.* A dung-recycling strategy has great potential for future farming. Farmers often use treated dung as a bedding material, but the safety of treated dung limits its application for calf farming. This study evaluated the feasibility of treated dung as a bedding material for suckling buffalo calves by comparing it with rice husk and rice straw bedding. We found that treated dung is feasible and optimal as bedding material for suckling buffalo calves. Our study provides guidance for manure recycling and calf farming.
https://doi.org/10.3168/jds.2022-22485.

Genetic relationships between weight loss in early lactation and daily milk production throughout lactation in Holstein cows. *By Tribout et al., page 4799.* After calving, high-yielding dairy cows mobilize their body reserves to produce

milk, resulting in weight loss and, in severe cases, health and fertility problems. This study uses daily milk production and live weight data recorded automatically by milking robots for 4,899 Holstein cows raised in commercial herds to estimate the genetic parameters and relationships between body weight loss in early lactation and milk yield throughout the lactation. Our results show that the genetic relationship between these 2 traits varies from moderate to very low, suggesting that it is possible to select for milk production without further increasing early body mobilization.
https://doi.org/10.3168/jds.2022-22813.

Validation of single-step genomic BLUP random regression test-day models and SNP effects analysis on milk yield in French Saanen goats. *By Arnal et al., page 4813.* In this study, 2 single-step genetic evaluation models were compared to determine which had the best ability to predict the genetic values of males at birth. Genomic information from the animals added to both models significantly improved the accuracy of the evaluations. The first model considered each monthly milk test day and the second considered the total milk production throughout lactation. The model based on test days reduced the overestimation of the evaluation and allowed for the selection of animals based on milk persistency (capacity to maintain production after peak milk production is reached).
https://doi.org/10.3168/jds.2022-22550.

Effects of type traits, inbreeding, and production on survival in US Jersey cattle. *By Nascimento et al., page 4825.* Survival is an economically important trait in dairy cattle. Here, we evaluated the effects of different type traits, inbreeding, and production level on the probability of survival in US Jersey cattle. Our results indicate that within-herd production level is the most important factor affecting the probability of survival, but some type traits, such as udder depth and high inbreeding values, also affect the probability of survival.
https://doi.org/10.3168/jds.2022-23048.

Improving national fertility evaluations by accounting for the rapid rise of embryo transfer in US dairy cattle. *By Miles et al., page 4836.* Commercial embryo transfer (ET) began in the 1970s and has become common practice in herds to increase their rate of genetic progress. Trends in ET-breeding event reporting do not parallel the ET calf birth rate, which can interfere with national genetic evaluations and on-farm fertility management. Completely censoring ET-associated records is not the desired approach, as these represent the most elite animals and herds, and unreported ET could bias fertility trait evaluations in the population. We describe an edit accounting for incorrect ET reporting and explore its effect on evaluations of sire, heifer, and cow conception rates.
https://doi.org/10.3168/jds.2022-22298.

Application of single-step single nucleotide polymorphism best linear unbiased predictor model with unknown-parent groups for type traits in Japanese Holsteins. *By Osawa et al., page 4847.* A single-step single nucleotide polymorphism best linear unbiased predictor model that combines phenotypes, pedigrees, and genotypes can be an option for routine genomic evaluation in dairy cattle. More reports are needed on the predictability of genomic evaluations and the computational feasibility of applying complicated statistical models with unknown-parent groups to compensate for the impact of missing pedigree on the bias of genomic evaluations. This study showed that the model with unknown-parent groups had little effect on the predictive ability and overdispersion of genomic predictions for young animals in Japanese Holsteins. Furthermore, the computation was not an issue if the solver was optimized.
https://doi.org/10.3168/jds.2022-22541.

Use of intravaginal lactic acid bacteria prepartum as an approach for preventing uterine disease and its association with fertility of lactating dairy cows. *By Madureira et al., page 4860.* Metritis is characterized as a uterine inflammation within 21 days of parturition and is one of the most prevalent bacterial diseases in dairy cows. The use of probiotic strains has been proposed to prevent postpartum uterine infections and inflammation. The combination of 3 strains of lactic acid bacteria (*Lactobacillus rhamnosus*, *Pediococcus acidilactici*, and *Lactobacillus reuteri*) applied during the 3 weeks before calving reduced the incidence risk of metritis. This, however, was not observed in both participating farms, suggesting that management could alter the benefits of this treatment. Probiotic treatment was not found to affect fertility overall on either farm, although an interaction with parity was detected, as well as an increase in cows found in estrus for the first artificial insemination postpartum. This study demonstrated that vaginal probiotic treatment applied 3 weeks before calving has the potential to decrease the incidence of metritis and, thus, an indirect influence on fertility.
https://doi.org/10.3168/jds.2022-22147.

Epidemiology of bovine colostrum production in New York Holstein herds: Cow, manage-

ment, and environmental factors. *By Westhoff et al., page 4874.* Adequate consumption of high-quality colostrum has been associated with reduced risk of preweaning calf morbidity and mortality. Nineteen New York Holstein dairy farms were enrolled to describe colostrum production and management and investigate the factors associated with the variation in colostrum production. Colostrum yield was associated with Brix %, calf sex, dry period length, previous lactation 305-day mature-equivalent milk yield and days in milk, gestation length, parity, prepartum environmental factors, and whether the calf was a stillbirth. Colostrum Brix % was associated with calf sex, dry period length, 305-day mature-equivalent milk yield of previous lactation, gestation length, colostrum yield, parity, and prepartum environmental factors, whether the calf was a stillbirth.
https://doi.org/10.3168/jds.2022-22447.

Epidemiology of bovine colostrum production in New York Holstein herds: Prepartum nutrition and metabolic indicators. *By Westhoff et al., page 4896.* Limited data describe the association of prepartum nutrition and energy metabolism with bovine colostrum production. Cows from 19 New York Holstein dairies were enrolled to identify herd-level nutritional strategies and cow-level metabolic indicators associated with colostrum production. Colostrum yield was associated with prepartum diet variables of starch, crude protein, dietary cation-anion difference, and particle length. Metabolic indicators of prepartum nonesterified fatty acid concentrations and herd-level postpartum hyperketonemia prevalence were also associated with colostrum yield. Colostrum Brix % was associated with prepartum diet variables of starch, neutral detergent fiber, dietary cation-anion difference level, and particle length.
https://doi.org/10.3168/jds.2022-22960.

Integrated meta-omics analyses reveal a role of ruminal microorganisms in ketone body accumulation and ketosis in lactating dairy cows. *By Wang et al., page 4906.* High-throughput sequencing and macrogenomic analysis revealed a strong correlation between β-hydroxybutyrate and the relative abundance of ruminal microorganisms and ketosis in dairy cows. Ruminal microbiota combined with blood metabolite analyses revealed that 4 microorganisms in the rumen promoted the development of ketosis by affecting fatty acid metabolism, acetyl coenzyme A, and ketone bodies, including β-hydroxybutyrate accumulation. Microorganisms such as *Succinivibrionaceae* were in lower abundance in the rumen of healthy cows than ketotic cows and are expected to have a regulatory role in the prevention of ketosis.
https://doi.org/10.3168/jds.2022-22282.

Leukocytes, microRNA, and complement activity in raw, heat-treated, and frozen colostrum and their dynamics as colostrum transitions to mature milk in dairy cows. *By Chandler et al., page 4918.* Colostrum is a critical source of immunoglobulins for bovine neonates, but comparatively less is known about other immunologically active factors that may affect immune development in calves. We investigated the effects of heat treatment and freezing on colostral leukocytes, microRNA, and complement activity as compared with raw colostrum, as well as their dynamics as colostrum transitions to mature milk. Cells were nonviable after both heat treatment and freezing; freezing increased microRNA abundance, and heat treatment decreased complement activity. Proportions of leukocyte populations differed between colostrum, transition, and mature milk, and microRNA abundance decreased in transition and mature milk.
https://doi.org/10.3168/jds.2022-22876.

An iterative approach to the development of a sole ulcer induction model in Holstein cows. *By Cramer et al., page 4932.* Induction models facilitate the study of health disorders by inducing the disorder under controlled, experimental conditions. These models serve as valuable tools in research aimed at developing and testing preventive and therapeutic treatment strategies. Developing a sole ulcer induction model would allow researchers to study the effects of different treatment and pain-mitigation strategies for sole ulcers, leading to improvements in cow health and welfare.
https://doi.org/10.3168/jds.2022-22726.

Effects of *Echinacea purpurea* supplementation on markers of immunity, health, intake, and growth of dairy calves. *By McNeil et al., page 4949.* The objective of this study was to investigate how supplementing calves with *Echinacea purpurea* affects markers of immunity, health, intake, and growth. *Echinacea purpurea* supplementation was associated with immunomodulation and reduced inflammation, evidenced through blood markers, with minor health and growth improvements. Benefits were observed, particularly when fed across the whole milk feeding period.
https://doi.org/10.3168/jds.2022-22862.

An observational study of ear-tagged calf mortality (1 to 100 days) on Irish dairy farms and associations between biosecurity practices and calf mortality on farms participating in a Johne's disease control program. *By McAloon et al., page 4966.* Postnatal mortality among replacement stock has a detrimental effect on the social, economic,

and environmental sustainability of dairy production. The objectives of this study were to quantify mortality in Irish dairy calves and determine risk factors for calf mortality over 5 years. Calf mortality rates in Irish dairy herds declined between 2016 and 2020. Our study suggests that implementing of recommended biocontainment practices to control paratuberculosis was associated with a reduction in calf mortality hazard.
https://doi.org/10.3168/jds.2022-22519.

Early detection of subclinical mastitis in lactating dairy cows using cow-level features. *By Pakrashi et al., page 4978.* There is a compelling need to predict the onset of subclinical mastitis in dairy cows sufficiently early to enable mitigation treatment. The present study focused on using data that could be recorded routinely at milking to accurately predict the onset of subclinical mastitis up to 7 days prior to onset. The most important cow-level attributes influencing predictive accuracy were milk somatic cell count, days since calving, parity, genetic merit, and the number of previous subclinical cases for that parity of that cow.
https://doi.org/10.3168/jds.2022-22803.

Evaluation of test-day milk somatic cell count to predict intramammary infection in late lactation grazing dairy cows. *By Clabby et al., page 4991.* This study evaluated test-day milk somatic cell count, parity, milk yield, and lactation history of high somatic cell count to predict intramammary infection in late lactation in 21 Irish dairy herds. We found the predominant pathogen responsible for intramammary infection was *Staphylococcus aureus* and a higher level of intramammary infection in primiparous than multiparous cows. Last test-day somatic cell count was the best predictor of intramammary infection in late lactation. The somatic cell count cut-point that maximized sensitivity and specificity was 65,000 cells per mL and should be explored as a decision-making tool for dry cow treatment selection on herds with predominantly *Staph. aureus* infections.
https://doi.org/10.3168/jds.2022-22627.

The effect of day-only versus day-plus-night cooling of dairy cows. *By Gaughan et al., page 5002.* Cooling dairy cows during peak daily heat is common practice. Cooling cows helps to alleviate environmental heat load and maintains cow well-being. However, the temperature gradient between the cow and the environment, especially when the ambient temperature is >35°C, limits the amount of heat that a cow with a body temperature ≥38.6°C can dissipate to the environment. This is especially problematic when nights do not cool, and cows' body temperatures remain elevated. Combining day cooling with night cooling may enhance the cooling effect, resulting in a faster return to normal body temperature, thereby enhancing well-being and productivity.
https://doi.org/10.3168/jds.2022-22819.

Association of time to metritis diagnosis with circulating concentration of metabolites, minerals, and haptoglobin in Jersey cows. *By Menta et al., page 5029.* The objective of this study was to evaluate the association of cows diagnosed with metritis at <5 days in milk (early metritis; EMET) and at ≥5 days in milk (late metritis; LMET) with circulating concentrations of metabolites, minerals, and haptoglobin. We also evaluated the effects of EMET and LMET on lactation performance compared with that of nonmetritic (NMET) cows. Haptoglobin concentrations were greater for EMET cows than for LMET and NMET counterparts, and EMET cows had a greater concentration of free fatty acids than NMET cows. However, no differences were observed in lactation performance.
https://doi.org/10.3168/jds.2022-22979.

Development and application of a novel approach to scoring ear tag wounds in dairy calves. *By Harmon et al., page 5043.* Ear tagging is a common practice in livestock management, but little is known about the animal welfare implications of the procedure. We developed a novel scoring system to quantify the wound-healing process after tagging dairy calves. Ear tags led to extensive damage in terms of tissue types (exudate, crust, tissue growth) and long-lasting wounds, which were present for at least 12 weeks. Piercings were also accompanied by wounds due to friction from the tag against the ear. Further research is warranted to understand the risk factors and ways to improve the practice.
https://doi.org/10.3168/jds.2022-23005.

Colostrum insulin supplementation to neonatal Holstein bulls affects small intestinal histomorphology, mRNA expression, and enzymatic activity with minor influences on peripheral metabolism. *By Hare et al., page 5054.* Neonatal bulls were fed colostrum that contained low (12.9 μg/L), moderate (70.0 μg/L), or high (149.7 μg/L) insulin concentrations at 2, 14, and 26 hours postnatal. Blood metabolites and insulin concentration were minimally affected by colostrum insulin concentration, whereas multiple effects were observed in intestinal development, indicating an improved surface area and absorptive capacity. Lactase activity linearly increased with greater colostrum insulin intake. Colostral insulin

affected intestinal development by 30 hours postnatal, leading to minor changes in peripheral metabolism.
https://doi.org/10.3168/jds.2022-22965.

Effects of source of supplementary trace minerals in pre- and postpartum diets on reproductive biology and performance in dairy cows. *By Mion et al., page 5076.* The complete replacement of inorganic sources of supplementary trace minerals by organic sources in both pre- and postpartum diets resulted in important differences in preimplantation conceptus development. These differences included major changes in gene expression that indicated an advanced conceptus development in cows supplemented with organic sources in addition to changes in the expression of genes directly related to cellular function of trace minerals. Modest benefits were also observed in copper concentrations in follicular fluid and time to resume estrous cyclicity, mostly in primiparous cows. Despite the effects on uterine and ovarian biology, no differences in reproductive performance were observed.
https://doi.org/10.3168/jds.2022-22784.

Abomasal infusion of essential fatty acids and conjugated linoleic acid during late pregnancy and early lactation affects immunohematological and oxidative stress markers in dairy cows. *By Veshkini et al., page 5096.* Hematologic and oxidative stress markers were evaluated in transition dairy cows supplemented with essential fatty acids, particularly α-linolenic acid, or conjugated linoleic acids. Immunohematological results revealed that dairy cows are in a state of oxidative stress immediately after parturition, which gradually subsides during the next few weeks. Oxidative stress markers were found to be correlated with inflammation markers. Fatty acid supplementation had minor effects on the markers of oxidative stress in a time-dependent manner. Significant differences were noted between the essential fatty acids and conjugated linoleic acids groups, which were related to hematologic markers and less so to oxidative stress.
https://doi.org/10.3168/jds.2022-22514.

Presynchronization with prostaglandin $F_{2\alpha}$ and gonadotropin-releasing hormone simultaneously improved first-service pregnancy per artificial insemination in lactating Holstein cows compared with Presynch-14 when combined with detection of estrus. *By Martins et al., page 5115.* Breeding strategies that combine detection of estrus and timed artificial insemination limit days in milk at first service and improve the reproductive performance of dairy farms. This study evaluated the effect of 2 simple breeding strategies on the fertility of the first service in lactating Holstein cows. The presynchronization strategy using $PGF_{2\alpha}$ and gonadotropin-releasing hormone simultaneously increased the proportion of cows receiving timed artificial insemination and fertility of first service than the presynchronization that only used $PGF_{2\alpha}$ (Presynch-14). Thus, $PGF_{2\alpha}$ plus gonadotropin-releasing hormone is a simple alternative breeding program that can be implemented in dairies using Presynch-14 to increase the first-service fertility of lactating Holstein cows.
https://doi.org/10.3168/jds.2022-22651.

AMPK-endoplasmic reticulum stress axis contributes to lipopolysaccharide-caused mitochondrial dysfunction by regulating mitochondria-associated membrane function in bovine hepatocytes. *By Xie et al., page 5146.* Our study examined whether endoplasmic reticulum (ER) stress contributed to mitochondrial dysfunction in lipopolysaccharide (LPS)-treated bovine hepatocytes. We found that inhibiting ER stress could reduce apoptosis and mitophagy to alleviate LPS-induced mitochondrial damage via inactivating adenosine monophosphate–activated protein kinase (AMPK). Inhibition of AMPK could also relieve ER stress and alleviate LPS-caused mitochondrial damage by mediating mitochondria-associated membrane dynamic function. Furthermore, we demonstrated that inhibiting special ER stress sensors (protein kinase–like ER kinase and inositol-requiring enzyme 1α) decreased levels of apoptosis and mitophagy and restored the balance of mitochondrial fission and fusion, indicating that protein kinase–like ER kinase and inositol-requiring enzyme 1α played an important role in regulating mitochondria-associated membrane activity to mediate mitochondrial function. Our research provides new insight into mitigating the negative effect of LPS on the livers of dairy cows.
https://doi.org/10.3168/jds.2022-22879.

Reproductive performance of lactating dairy cows with elevated milk β-hydroxybutyrate levels during first 6 weeks of lactation. *By Alemu et al., page 5165.* In this study, we evaluated the reproductive performance of cows with elevated milk β-hydroxybutyrate, indicative of ketosis, during 2 periods of early lactation. The results of the study were based on data from more than 30,000 cows and demonstrated that there is a negative association between elevated milk β-hydroxybutyrate within 42 days in milk and reproductive performance after the voluntary waiting period, over and above other determinant factors such as parity, production level, and season. Although cows suspected of elevated milk β-hydroxybutyrate during 5 to 14 days in milk did not have an adverse effect on their reproduc-

tive performance, cows suspected of elevated milk β-hydroxybutyrate during 15 to 42 days in milk had significantly reduced reproductive performance. https://doi.org/10.3168/jds.2022-22406.

Tumor necrosis factor-α reduces adiponectin production by decreasing transcriptional activity of peroxisome proliferator-activated receptor-γ in calf adipocytes. *By Yu et al., page 5182.* Adiponectin increases insulin sensitivity in dairy cows, and its plasma concentration declines at parturition. Adipose tissue inflammation, a common finding in transition dairy cows, increases production of proinflammatory cytokines. We found that tumor necrosis factor-α (TNF-α) decreased transcriptional activity of peroxisome proliferator-activated receptor-γ (PPARγ), downregulated expression of adiponectin assembly regulators, and reduced adiponectin secretion in calf adipocytes. Knockdown and overexpression of PPARγ, respectively, suppressed and enhanced adiponectin transcription, assembly, and secretion. In TNF-α-treated calf adipocytes, PPARγ overexpression increased adiponectin production. Thus, elevated TNF-α levels in adipose tissues may partly reduce adiponectin assembly by inhibiting PPARγ transcriptional activity in periparturient cows. https://doi.org/10.3168/jds.2022-22919.